

Orion Hills Observatory

Research Report 2025

Orion Hills Observatory - Eastern Oregon, United States

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two state universities.

and modernised in 2014. The observatory is operated by a nonprofit research consortium together with The main instrument is the Pioneer telescope, a 1.8 metre Cassegrain reflector commissioned in 1979 decade.

metres. The site was selected for its number of clear nights, which averaged 212 per year during the last Orion Hills Observatory was founded in 1963 on a ridge in eastern Oregon at an altitude of 2,140

Institution Overview

events every clear night.

degrees. A robotic 0.4 metre patrol telescope installed in 2021 screens nearby star fields for transient

The wide-field camera Lattice uses a mosaic of twelve CCD chips and covers a field of 1.7 square

resolving power of $R = 42,000$, fed by an optical fibre from the Cassegrain focus.

Besides the Pioneer telescope, the observatory operates the Vesper echelle spectrograph with a

Instrumentation

candidates were confirmed as planets by radial velocity measurements with the Vesper spectrograph. The exoplanet transit program observed 3,100 target stars and registered 17 transit candidates. Four numbers during the year. Follow-up astrometry is shared with four amateur stations. The asteroid survey discovered 1,342 new minor planets in 2025, of which 87 received permanent

Survey Programs

days. All light curves are published within one year of observation. In 2025 the program added 44 newly discovered eclipsing binaries with periods between 0.9 and 42 taken with the Lattice camera in two filters, and light curves reach a period precision of 0.4 percent. A long term monitoring catalogue covers 386 RR Lyrae stars in the northern sky. Observations are

Variable Star Monitoring

million USD.

the survey programs, followed by university contributions of 1.5 million USD and private donations of 1.1 million USD. The annual budget was 3.8 million USD. The largest single item is a federal grant of 1.2 million USD for the American Astronomical Society family.

Staff and visiting astronomers published 21 peer reviewed papers in 2025, twelve of them in journals of

Publications and Funding

pipeline.

reduced real survey data and contributed 14 asteroids measurements to the minor planet center

A summer academy for 30 undergraduate students ran for eight weeks in July and August. Participants

daytime visits, and monthly public star parties were held from April to October.

The visitor programme counted 9,400 guests in 2025. Sixty-two school classes took part in guided

Education and Outreach